

GaN-Based Low-Voltage Inverter for Electric Scooter Drive System

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Abstract— The paper deals with the application of the latest generation low voltage (up to 80V) Gallium Nitride (GaN) devices for motor drive applications in the field of electric micro-mobility. A two-level inverter topology oriented to light electric traction (such as electric scooters) with a power rate of 1500W has been considered. An experimental board using two GaN FETs in parallel connection for each switch of an inverter leg has been arranged. The intrinsic parameters spread of the device, as well as the leakage inductances of both the circuit and the PCB, to evaluate the impact on the current share during both the transient phase and in the steady state have been investigated. In the paper, several simulation runs have been carried out and described. Furthermore, the inverter experimental board has been used to evaluate the phase voltages and currents at different load conditions. Finally, the temperature limits versus phase current variation have been evaluated.

Keywords—GaN FET, Inverter, Parallel Connection, Micro-mobility.

I. INTRODUCTION

In city transport, electric scooters (e-Scooter) are affordable for sustainable mobility with zero CO₂ emissions, drivability with heavy traffic, and parking in a limited space. E-Scooters belong to the micro-mobility market which is expected to reach 214.57 billion \$ by 2030 [1]. Micro-mobility is a variegated and not yet standardized electric mobility scenario, made up of several relatively small battery-powered vehicle systems with autonomy ranges depending on many factors, such as weight, maximum torque, speed, number of people to move, safety, and cost [2].

Light electric vehicles (EV) such as E-kick scooters, e-Bikes, Skateboards, Hoverboards, Low-Speed EVs, Segways, Mopeds, and e-Scooters are some examples that we can currently find in our cities both as private vehicles as well as in the new sharing availability. A general block schematic of battery storage and electric propulsion system for Light EVs is depicted in Fig. 1. Nowadays, the battery storage pack solution in several applications for micro-mobility is composed of Lithium-based batteries [3]. The battery storage pack is managed by the Battery Management System (BMS). The voltage rate of the battery pack is in the range of 24 V, up to 120 V. The Energy rate average capacities are between 150 Wh and 750 Wh, but some expensive scooters and disabled-people four-wheel e-Scooters feature storage battery packs with much more energy density.

The propulsion system is composed first of an electric motor, generally a DC brushless for a strictly economic or permanent magnetic sinusoidal motor (PMSM) for more

qualitative movement applications [4]. The power electronic converter is a two-level three-phase inverter topology. The switching modulation strategy is driven by an electronic control unit (ECU) linked with the BMS and the motor parameters feedback to obtain the torque and speed control requested.

The technology of the power switches is crucial to obtain an efficient and reliable inverter system. The pure Silicon MOSFETs are currently used in the typical power range of hundreds of Watt up to a few kW for storage battery pack voltage rate below 200 V [5]. In recent years, High Electronic Mobility Transistor (HEMT) based on Gallium Nitrate (GaN) is introduced in application involving the considered range of power and voltage rate. The GaN FET devices are more attractive with several enhancements compared with the low voltage new generation of pure silicon trench-gate MOSFET [6].

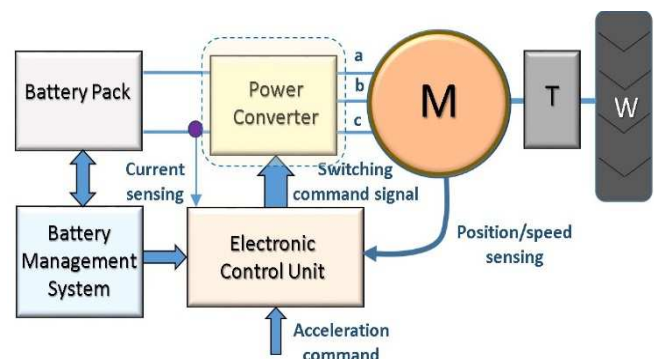


Fig. 1. Electric Storage and Propulsion scheme of Light Electric Vehicles such as an e-Scooter. Where M is a generic electric motor, T is the transmission box, and W is the moving wheel.

In the paper, a 60 V, 1500W two-level inverter topology based on low voltage GaN FET is evaluated. The inverter topology solution is suitable for a high-efficiency inverter for a high-performance e-Scooter drive application. To reach the switching leg current requested (35 Arms), two devices are paralleled for each high-side and low-side legs switches. It is worth investigating the parallel connection of GaN FET devices to understand how to obtain switches with a high operating current. The paper is structured as described in the following. In the second section, a parallel connection and the link with the technological issues are explored.

In the third section, a simulation investigation in LTSpice is carried out to evaluate the current and voltage sharing in the two paralleled connected GaN devices. Furthermore, the circuit and device stray inductances are considered.

In the fourth section, an experimental evaluation is presented and discussed.

Finally, in the last section, the conclusion and the future works are highlighted.

II. PARALLEL CONNECTION AND TECHNOLOGIC ISSUES OF GAN FET DEVICES

The GaN FET switches considered are enhancement-mode devices. The HEMT is a planar structure with high electron mobility due to a two-dimensional electron gas (2DEG) found at the interface between an AlGaIn and GaN heterostructure interface [7]. The GaN FET has a p-type GaN layer used to make a normally-off enhancement-mode device. In Fig. 2a the structure of an enhancement-mode GaN FET is depicted. The devices considered in this paper feature a breakdown voltage until 200V. The low voltage applications at high switching frequency (until tens of MHz) are the typical field of operation in high-performance DC-DC or DC-AC converters [8], [9]. The GaN FET is a bidirectional device as a MOSFET one [10]. The operation in the first and third quadrant is obtained by a positive voltage polarization of the Gate-Source (5 V). At zero voltage across Gate-Source, the reverse conduction in a GaN FET is similar to a diode with a voltage drop higher than a body diode of a MOSFET [11]. In Reverse conduction, the equivalent diode of the GaN FET does not show a reverse recovery charge Q_{rr} . Encapsulation in the device package plays a key role in achieving low leakage inductances. The structure of the actual package with Solder Bars is reported in Fig. 2b. The GaN FETs are high-density current devices with a direct conduction resistance $R_{DS(on)}$ with a positive temperature coefficient (Fig. 2c) like a power MOSFET [7].

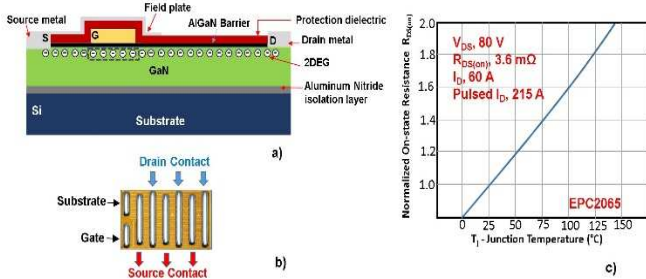


Fig. 2. a) Physical structure of an enhancement-mode GaN FET. b) Packed GaN FET with the solder bars in the low side device. c) Normalized $R_{DS(on)}$ for an EPC2065 GaN FET versus the temperature at $I_D=25\text{ A}$, $V_{GS}=5\text{ V}$.

The positive temperature coefficient allows a viable way to realize a high current switch by the parallel connection of several GaN FETs. Generally, the current imbalance in parallel connection is caused by the device parameters mismatch and the printed circuit board (PCB) design choices [6]. The $R_{DS(on)}$ spread affects the on-state current distribution among the parallel-connected switches. Furthermore, the thermal differences between the devices lead to a different on-state current sharing too. The paralleled devices should be in similar thermal conditions with the same thermal resistance and in a common heat-sink arrangement. In switching operations, the trans-conductance (g_m) influences the turn-on and turn-off dynamic current sharing. However, in insulated gate devices, the main parameter involved in the current difference during the transient behavior is the threshold voltage V_{TH} as discussed in [5]. A lower V_{TH} results in an

earlier turn-on featuring a high current peak with consequently high switching power loss. A V_{TH} typical value at 5 mA is equal to 1.4V. Furthermore, the threshold voltage is relatively constant when increasing temperature, while in Si MOSFET the V_{TH} curve declines more rapidly [7]. Further factors affecting the parallel connection are the device package parasitic inductance and PCB layout parasitic inductance. The device package inductances are related to the package solution. As described in Fig. 2b, the device package features several drains and source paths to reduce the stray inductance. Furthermore, Chip Scale Package (CSP) and quad-flat no-leads (QFN) package solution lead to a suitable reduction in the parasitic inductances. These effects will be deeper described in the next section.

III. CURRENT SHARING EVALUATION IN PARALLELED GAN FET DEVICES

In this section, the problems that arise when connecting more GaN FETs in parallel are investigated. To achieve the purpose of the paper, several simulations are carried out using the LTspice software. In this work, a symmetrical topology consisting of two GaN FET connected in parallel, named A and B, is considered. It is shown in Fig.3.

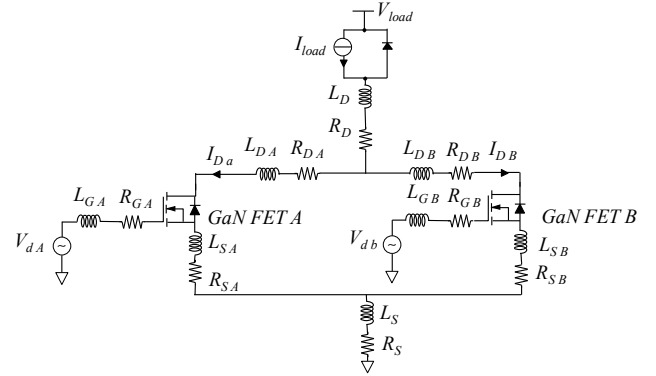


Fig. 3. Two GaN FETs in parallel connection with symmetrical arrangement. Parasitic impedances of connection lines are considered.

In Fig.3 the parasitic impedances of the gate R_G L_G , of the source R_S L_S , and of the drain R_D L_D are indicated with the subscript $_A$ or $_B$ depending on which device they refer to. No subscripts are used for indicating the common impedance at the source and the drain of the connected devices. R_G indicates the gate resistance and its value is $10\ \Omega$ for every GaN FET. The value of the gate resistance is chosen to have a slew rate drop (95%-5%) of 5 V/ns which is good for a motor drive application.

The GaN FET model used in the simulations is obtained from the basic model of a typical MOSFET, and it has been modified to best represent the device operation, including its internal parasitic effects [12].

The driving signals are sent synchronously to every device. The driving voltage V_d has been set as a 5 V square waveform voltage source with rise time and fall time equal to 0.1 ns and with a duty cycle of 0.5. The switching frequency is 10 kHz, but GaN FET technology can work with a switching frequency up to 1 MHz. Anyway, in this study, the value of the switching frequency is irrelevant because the commutation transients are not related to its value. The load is modeled as a current source I_{load} . It provides 40 A, and according to that, each GaN FET conducts a drain current of 20 A. The power

system works with a DC load voltage of $V_{load} = 48\text{ V}$. An ideal freewheel diode is connected in parallel with the load current source.

At first, simulations to study the effects of parasitic impedances of the PCB are presented. Subsequently, the study continues with simulations focusing on the effects caused by parameter spreads between GaN FETs connected in parallel. These effects are analyzed one at a time individually.

A. Effects induced by PCB parasitic components

The first parasitic components of the PCB to be studied are the common source impedances. In the circuit in Fig.4, only the parasitic source components are considered and they have values $L_{Sn} = 7.5\text{ nH}$, $R_{Sn} = 5\text{ }\mu\Omega$, $L_S = 6\text{ nH}$, $R_S = 5\text{ m}\Omega$. Due to the symmetry of parallel devices considered, all waveforms of electrical variables are equal between devices. Therefore, those of a generic n^{th} GaN FET are considered.

Table 1 reports the slow rates $\Delta I_{Dn}/\Delta t$ from 10% to 90% of the conduction value of the drain current and the duration of the drain current commutation in the cases with a non-null and null common source impedance Z_{Sn} .

Z_{Sn}	$\frac{\Delta I_{Dn}}{\Delta t}$ turn-on	Δt [ns] turn-on	$\frac{\Delta I_{Dn}}{\Delta t}$ turn-off	Δt [ns] turn-off
$\neq 0$	$136.46 \cdot 10^6$	148	$-112.66 \cdot 10^6$	180
$= 0$	$6.343 \cdot 10^9$	6	$-6,511 \cdot 10^9$	5

Table 1 Slow rates $\pm 10\%$ and duration of the drain current transients: with and without common source impedance Z_{Sn}

The main effect of the common source impedance is to slow down the drain current transition. This means that the whole switching process will last longer and the energy dissipated increases.

A simulation considering the two GaN FET in parallel connection and having both an equal parasitic gate inductance is performed. Typically, the leakage inductances can be caused by the length of the gate wires. The values of $L_{Gn} = 10\text{ nH}$ and $R_{Gn} = 0.1\text{ }\Omega$ are considered. The waveforms of the drain current, the gate-source voltage, and the drain-source voltages of a GaN FET in parallel connection with parasitic source impedance are shown in Fig.4. The turn-on and turn-off commutations are depicted in Fig.4a and Fig.4b respectively.

Fig.4 shows that there is the phenomenon of gate ringing. It results from the connection of gate inductance to the capacitance forming a resonant circuit. During the turn-off commutation, a false turn-on happens when the gate-source voltage surpasses the threshold level. To avoid this phenomenon L_G/L_S must be equalized and kept as low as possible. Using a proper gate resistance to source resistance ratio (R_G/R_S) reduces the gate ringing phenomenon, dampening the system. To avoid the ringing phenomenon, the gate resistance value must be bigger than the value expressed in (1) [7].

$$R_G = 2 \cdot \sqrt{\frac{L_S + L_G}{C_{GS}}} \quad (1)$$

Furthermore, the operative condition of two GaN FETs in parallel connection with only parasitic drain impedance is considered. The parasitic component values used in the simulation are $L_{Dn} = 0.1\text{ nH}$, $R_D = 5\text{ m}\Omega$, $L_D = 0.1\text{ nH}$

Again, the symmetrical topology of the PCB considered as a hypothesis makes the current and voltage waveforms of devices equal. The waveforms of the drain current, the gate-source voltage, and the drain-source voltages of a GaN FET in parallel connection with parasitic drain impedance are shown in Fig.5a and Fig.5b for turn-on and turn-off respectability. Similarly, in the case of parasitic gate impedance, a ringing phenomenon occurs. It is most evident during the turn-off commutation. The inductances in the power loop generate higher frequency current and drain voltage fluctuations. They increase the voltage overshoot across the device and may cause voltage breakdown of the device. These are damped by the parasitic resistance and in the considered case they cancel out in approximately 140 ns. Differently from the parasitic gate impedance case, the gate-source voltage does not differ from the ideal situation without parasitic components. To reduce the ringing phenomenon, a reduced value of power loop leakage inductance must be achieved by an accurate layout design of the PCB.

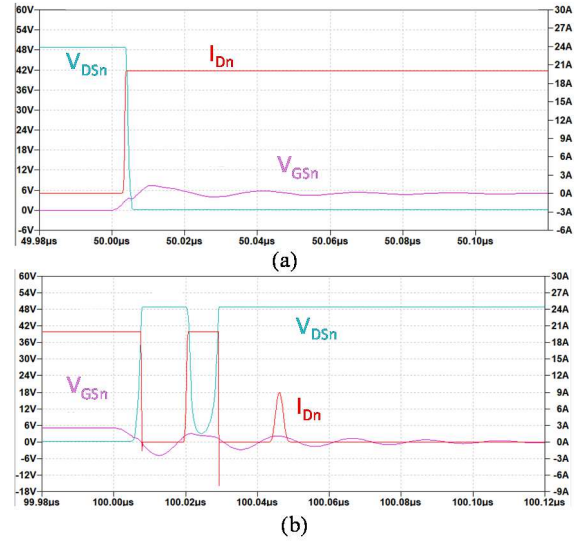


Fig. 4. Drain current in red, drain-source voltage in light blue, and gate-source voltage in magenta of a generic n GaN FET connected in parallel with parasitic gate impedance – low R_G case: (a) turn-on ; (b) turn-off. Time step=20 ns/div, $I=3\text{ A/div}$, $V=6\text{ V/div}$.

B. Effects induced by parametric spread of parallel-connected devices

Simulations where two EPC2065 (furnished by Efficient Power Conversion - EPC®) GaN FETs, named A and B, having a spread of one parameter are connected in parallel are carried out. No parasitic components are present in the topology of the system. The first parameter spread to be analyzed is the value of the device gain. One device may have a transconductance gain that is slightly different from that of another device. For example, this can be caused by manufacture mismatching, because materials coming from different batches are used to make the switching devices. Transconductance gain is defined as

$$g_m = dI_D/dV_{GS} \quad (2)$$

Acting on the GaN model parameter, device A has been modified to have a 10% reduced gain compared to device B. Both devices are driven with the same drive voltage to supply a 40 A load current. The gate-source voltage waveform

remains congruent for both GaN FETs, but a difference in drain current value is present.

The waveforms of the drain currents carried by the A and B devices and their gate-source voltage are depicted in Fig.6a for the turn-on and in Fig.6b for the turn-off. The gain spread makes the current of the GaN FET B with higher gain to show rise and fall times steeper than GaN FET A during commutations. However, no huge current peak phenomena are experienced during commutations because both devices reach the conduction current value (after the turn-on) and null current values (after turn-off) at the same time. During the conduction phase, a current spread between the two devices is present. It is proportional to the difference in gain.

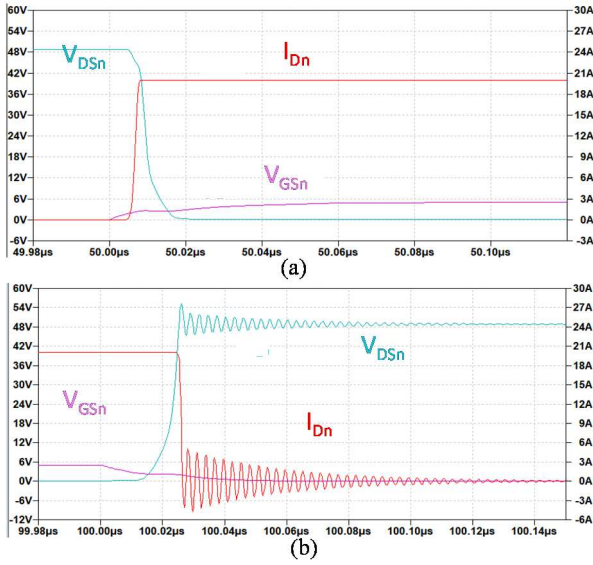


Fig. 5. Drain current in red, drain-source voltage in light blue and gate-source voltage in magenta of a generic n GaN FET connected in parallel with parasitic drain impedance: (a) turn-on ; (b) turn-off. Time step=20 ns/div, I=3 A/div, V=6 V/div.

The conduction resistance $R_{DS(on)}$ determines the voltage drop across the device and in GaN FET technology its value increases proportionally with junction-to-ambient temperature. The operative condition in which the GaN FET B has the typical value of conduction resistance $R_{DS(on)} = 2.2 \text{ m}\Omega$ and the GaN FET A has a grown +20% $R_{DS(on)}$ value is investigated. Fig.7 shows the waveforms of the gate-source voltage and the drain current provided by the two GaN FET in parallel connection. The zooms on turn-on and turn-off commutations are depicted in Fig.7a and Fig.7b respectability. The spread of the conduction resistance influences the current distribution only in the conduction time interval. The device with lower resistance (B) conducts more current. The current spread is proportional to the load current amplitude and the difference in conduction resistance of the two devices (it is 2 A in the considered case). However, no differences in the drain currents are present during commutations. If the spread is too large, joule losses could cause the device to heat up beyond the maximum permissible junction temperature. Likely the MOSFET thermal behavior, GaN FET conduction resistance has a positive temperature coefficient. It causes the hottest device to provide less current and decrease its losses.

Thus, the temperature coefficient trend equilibrates the $R_{DS(on)}$ and drain current spreads.

Threshold voltage V_{TH} is the minimum gate-source voltage needed to turn on enhancement mode transistors. Fig.8 shows the gate-source voltage and the drain current waveforms of two parallel-connected GaN FETs. GaN FET A has $V_{THA} = 0.7 \text{ V}$ and GaN FET B has $V_{THB} = 1.7 \text{ V}$. These values come from a $\pm 0.5 \text{ V}$ spread from the typical value declared of manufacturing. Fig.8a and Fig.8b show the turn-on and turn-off commutations respectability. When devices are connected in parallel, their V_{TH} spread leads the GaN FET with lower threshold voltage to be turned on earlier and turned off later (GaN FET A in this example). Looking at Fig.8, the parallel connection makes the gate-source voltage waveforms almost congruent.

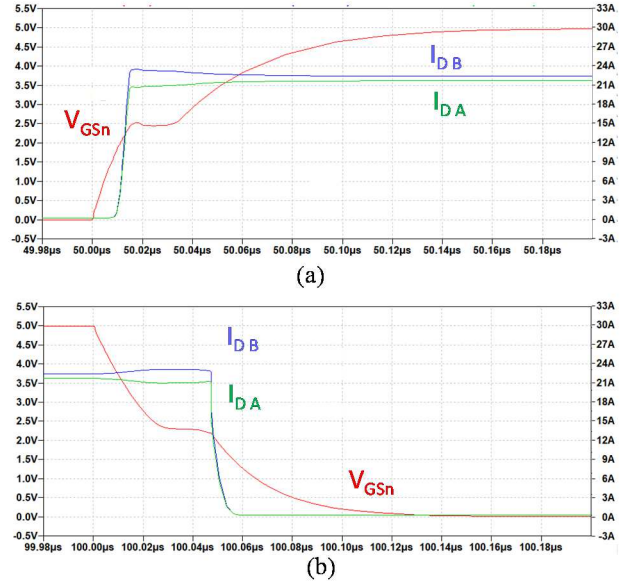


Fig. 6. Gate-source voltage of A and B GaN FETs in red, drain current of the A GaN FET in green, drain current of the B GaN FET in blue – $g_m(A)=90\%$ $g_m(B)$: (a) turn-on ; (b) turn-off. Time step=20 ns/div, I=3 A/div, V=0.5 V/div.

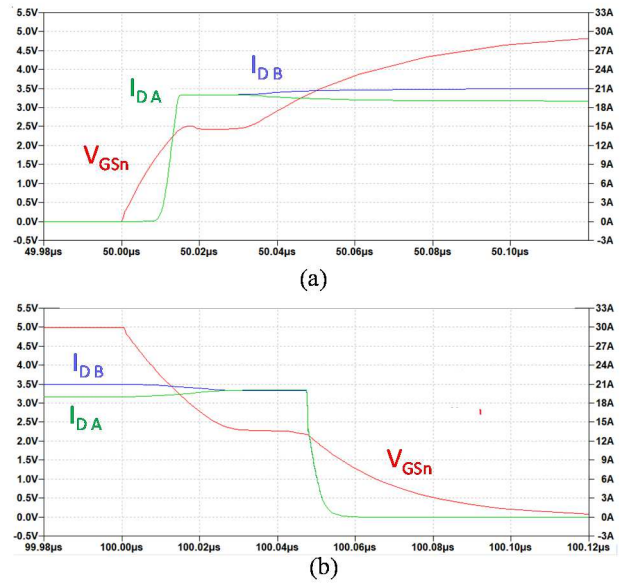


Fig. 7. Gate-source voltage of A and B GaN FETs in red, drain current of the A GaN FET in green, drain current of the B GaN FET in blue – in blue – $R_{DS(on)}(A) = 120\%$ $R_{DS(on)}(B)$: (a) turn-on ; (b) turn-off. Time step=20 ns/div, I=3 A/div, V=0.5 V/div.

Figure 10 consists of two subplots, (a) and (b), showing the measured gate voltages and drain currents of the 1T1R device during read operations. Both plots share a common x-axis representing time in microseconds (μs), ranging from 49.98 μs to 50.12 μs for (a) and 99.98 μs to 100.12 μs for (b). The left y-axis represents voltage in Volts (V), ranging from -0.5V to 5.5V. The right y-axis represents current in Amperes (A), ranging from -4A to 44A.

Subplot (a) shows the first read operation. The gate voltages V_{GSB} (green line) and V_{GSA} (red line) are plotted. V_{GSB} rises sharply from 0V to approximately 5.0V at 50.00 μs and then decays. V_{GSA} rises from 0V to approximately 2.5V at 50.00 μs and then decays. The drain currents I_{DA} (green line) and I_{DB} (blue line) are also plotted. I_{DA} rises sharply from 0A to approximately 40A at 50.00 μs and then decays. I_{DB} rises from 0A to approximately 2.0A at 50.00 μs and then decays.

Subplot (b) shows the second read operation. The gate voltages V_{GSB} (green line) and V_{GSA} (red line) are plotted. V_{GSB} rises sharply from 0V to approximately 5.0V at 100.00 μs and then decays. V_{GSA} rises from 0V to approximately 2.5V at 100.00 μs and then decays. The drain currents I_{DA} (green line) and I_{DB} (blue line) are also plotted. I_{DA} rises sharply from 0A to approximately 40A at 100.00 μs and then decays. I_{DB} rises from 0A to approximately 2.0A at 100.00 μs and then decays.

IV. EXPERIMENTAL EVALUATION OF A PARALLELED GAN BASED INVERTER

A. Temperature Evaluation

$$T = \left(\frac{V_{Temp} \cdot 1000}{7.87} \right) - 273.16 [^{\circ}C] \quad (3)$$

The graph shows a linear relationship between the Temperature Sensor T (°C) and the GaN FET Temperature T (°C). The x-axis is labeled 'Temperature Sensor T (°C)' and ranges from 20 to 120. The y-axis is labeled 'GaN FET Temperature T (°C)' and ranges from 20 to 120. A blue line starts at approximately (35, 38) and ends at (100, 100).

Temperature Sensor T (°C)	GaN FET Temperature T (°C)
35	38
40	45
50	55
60	65
70	75
80	85
90	95
100	100

The thermal performance of the inverter board is depicted in Fig. 11. The motor drive (BLDC motor) is designed for a 48 V_{DC} supply and operates with and without heatsink at natural convection. The tested phase current is 25 A_{RMS} at two switching frequencies (50kHz and 100kHz). The temperature increase (ΔT) compared with the ambient temperature ($T_{amb}=25.5\text{ }^{\circ}\text{C}$) is reported in Fig. 11 versus the phase current amplitude variation.

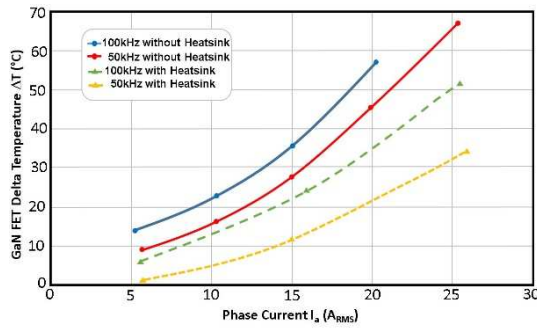


Fig. 11. The ΔT temperature versus the current variation at 50 kHz and 100 kHz without and with heatsink.

B. Switching operation

The switching results regarding the output voltage and current of the phase a of the inverter circuit (Fig. 9). The experimental tests have been carried out at $V_{dc}=48$ V and $I_{apeak}=50$ A. In Fig. 13 both the output current and voltage of the phase a are reported. In Fig. 12a the phase current I_a and the modulation voltage V_{am} are reported at motor frequency operation. In the zoomed view the switching phase voltage V_a and the phase current I_a are shown.

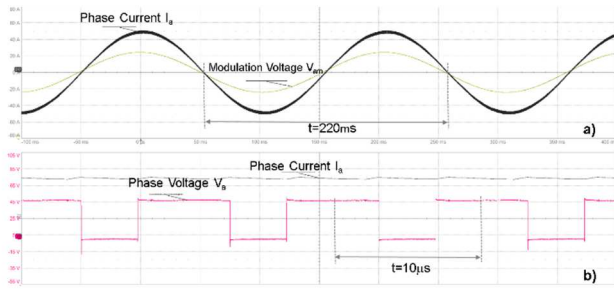


Fig. 12. Experimental waveforms at $V_{dc}=48$ V and $I_{apeak}=50$ A, $f_{sw}=100$ kHz. $V_a=20$ v/div, $I_a=20$ A/div, a) $t=50$ ms/div, b) zoomed view $t=10$ μ s.

In Fig. 13 the switching transient at turn-off (Fig. 13a) of the phase voltage (phase a) is shown with the current I_a at the peak value. The turn-on voltage (V_a) transient is reported in Fig. 13b.

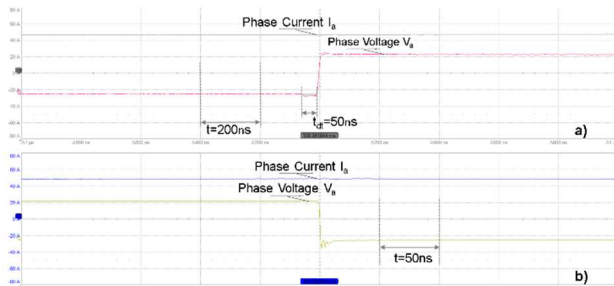


Fig. 13. Phase voltage transient waveforms at $V_{dc}=48$ V and $I_{apeak}=50$ A, $f_{sw}=100$ kHz, dead-time $t_{dt}=50$ ns. $V_a=20$ v/div, $I_a=20$ A/div, $t=50$ ns/div.

V. CONCLUSIONS

In the paper, a two-level low-voltage inverter topology for light electric traction (such as electric scooter) based on GaN FET switches in parallel connection has been evaluated and discussed. The dynamic current mismatching is mainly caused by the difference in V_{TH} . Several experimental tests concerned

the thermal behavior at different switching frequency are carried out. Finally, the switching frequency behavior of the output voltage and current related at one phase at a maximum rated current of 50 A are reported. In future work, a high current switch with several GaN FETs in parallel connection will be investigated.

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